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Ordinary Differential Equation Notes

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Chapter 1

First Order Differential Equations

For the first order differential equations, we would consider to solve

$$\frac{\mathrm{d}y}{\mathrm{d}t} = f(t, y)$$

Consider a simpler case:

$$\frac{\mathrm{d}}{\mathrm{d}} = g(t)$$

One with elementary calculus knowledge can solve it:

$$y(t) = \int g(t)dt + c$$

However, for general first order equation, it's very hard to solve.
We would first consider some simpler cases.

1 First order linear differential equations

Consider f is consisted by linear equations.

Definition 1.1.1 (First-oder linear differential equation)**.**

The general first-order linear differential equation is

$$\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = b(t) \tag{1}$$

Needless to say, $a(t)$, $b(t)$ are assumed to be continuous functions of time. To slove general first order linear equation(1), we could consider a simpler case:

Definition 1.1.2 (Homogeneous first order linear equation)**.**
The equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = 0 \tag{2}$$

is called homogeneous first-order linear differential equation

Now we try to solve them part by part.

1.1 Homogeneous first order linear differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = 0.$$

$$\frac{\mathrm{d}y}{\mathrm{d}t} = -a(t)y.$$

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}t} = -a(t).$$

Integrating both sides with respect to t gives

$$\int \frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}t}\,dt = -\int a(t)\,dt.$$

By the chain rule, $\frac{1}{y}\frac{dy}{dt} = \frac{d}{dt}\ln|y|$, hence

$$\ln|y(t)| = -\int a(t)\,dt + C,$$

where C is an arbitrary constant of integration. Exponentiating both sides yields

$$|y(t)| = e^{C}e^{-\int a(t)\,dt}.$$

Since $e^C > 0$, we may absorb the sign into a new constant $c = \pm e^C$, obtaining the general solution

$$\boxed{y(t) = c\,e^{-\int a(t)\,dt}} \tag{3}$$

where $c \in \mathbb{R}$ is an arbitrary constant. Note that the zero solution corresponds to $c = 0$.

In most cases, we are not interest in all solutions of (3). We could add more assmup-tion as

$$y(t_0) = y_0 \tag{4}$$

Then we integral (3) between t_0 and t.
Recalling $\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}t} = -a(t)$, we could write:

$$\int_{t_0}^{t} \frac{\mathrm{d}}{\mathrm{d}s}\ln|y(s)|ds = \int_{t_0}^{t} -a(s)ds$$

So

$$\ln|y(t)| - \ln|y(t_0)| = \ln|\frac{y(t)}{y(t_0)}| = -\int_{t_0}^{t} a(s)ds$$

Thus,

$$|\frac{y(t)}{y(t_0}| = \exp(-\int_{t_0}^{t} a(s)ds)$$

The problem now becomes +1 or -1. By taking $t = t_0$, we can find it's +1.
So

$$y(t) = y(t_0)\exp(\int_{t_0}^{t} a(s)ds) \tag{5}$$

1.2 General first order linear differential equation

Consider the problem:

$$\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = b(t) \tag{6}$$

We want to write it in a form:

$$\frac{\mathrm{d}'something'}{\mathrm{d}t} = c(t)$$

and then we can solve it. Multiply both sides by a continuous function $\mu(t)$ [1]
So

$$\mu(t)\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)\mu(t)y = \mu(t)b(t)$$

We consider another formula: $\frac{\mathrm{d}}{\mathrm{d}t}\mu(t)y = \frac{\mathrm{d}y}{\mathrm{d}t}\mu(t) + y\frac{\mathrm{d}\mu(t)}{\mathrm{d}t}$.
So we try to choose proper μ s.t. the left of the equation can be written as $\frac{\mathrm{d}'something'}{\mathrm{d}t}$.
We choose

$$\mu(t) = \exp(\int a(t)dt) \tag{7}$$

And

$$\frac{\mathrm{d}}{\mathrm{d}t}\mu(t)y = \mu(t)b(t)$$

$$\mu(t)y = \int \mu(t)b(t)dt + c$$

$$y = \frac{1}{\mu(t)}(\int \mu(t)b(t)dt + c), \quad \text{with } \mu(t) = \exp(\int a(t)dt) \tag{8}$$

We could consider $y(t_0) = y_0$ So:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mu(t)y = \mu(t)b(t)$$

$$\Downarrow$$

$$\mu(t)y - \mu(t_0)y_0 = \int_{t_0}^{t} \mu(s)b(s)ds$$

Thus,

$$y = \frac{1}{\mu(t)}\Big(\mu(t_0)y_0 + \int_{t_0}^{t} \mu(s)b(s)dt\Big) \tag{9}$$

[1] Here any continuous μ could satisfies the condition.

2 Separable equations

Recall how we deal with linear differential equation:

$$\frac{1}{y(t)}\frac{\mathrm{d}y}{\mathrm{d}t} = -a(t)$$

$$\Downarrow$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\ln|g(t)| = -a(t)$$

So for a more general ODE with the form:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{g(t)}{f(y)} \tag{10}$$

We could write it as

$$\frac{\mathrm{d}}{\mathrm{d}t}F(y(t)) = g(t), \quad \text{with } F(y) = \int f(y)dy$$

So the answer is:

$$F(y(t)) = \int g(t)dt + c \tag{11}$$

We could also add an assumption $y(t_0) = y_0$
So

$$F(y(t)) - F(y(t_0)) = \int_{t_0}^{t} g(s)ds$$

$$\int_{y_0}^{y} f(r)dr = \int_{t_0}^{t} g(s)ds$$

3 Population models

Let $p(t)$ denotes the population of a given species at time t, $r(t, p)$ denotes the difference between its birth rate and death rate.
For the simplest, we think that r will not change with t and p, which means it's a constant. The simplest model:

$$\frac{\mathrm{d}p(t)}{\mathrm{d}t} = ap(t), \quad a \text{ is a constant}$$

Also, assume $p(t_0) = p_0$
We can easily find the solution

$$p(t) = p_0 \exp(a(t - t_0)) \tag{12}$$

This result works very well if the population is small. However, if the population grows larger and larger, it falls. So we can let

$$\frac{\mathrm{d}r}{\mathrm{d}t} = ar - br^2 \tag{13}$$

It's also a sparable problem, though the integer part is a bit complex (but absolutly solvable!)
The answer is:

$$p(t) = \frac{ap_0 e^{a(t-t_0)}}{a - bp_0 + bp_0 e^{a(t-t_0)}} = \frac{ap_0}{bp_0 + (a - bp_0)e^{-a(t-t_0)}}.$$

4 Exact equations and the range of equations we can solve

Consider all differential equation we've solved, we can put them in this form:

$$\frac{\mathrm{d}}{\mathrm{d}t}\phi(t,y) = 0 \tag{14}$$

Example 1.4.1 (ϕ for first order linear differential equations)**.**
Thinking about all sparable differential equation (it would also contains all first order linear differential equations) we've solved before:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = g(t)h(y(t))$$

We can all write this equation in this form:

$$\phi(t,y) = \int^y \frac{1}{h(s)}ds - \int^t g(r)dr$$

More generally, we can solve all differential equations which can be written in this form:

$$\frac{\mathrm{d}}{\mathrm{d}t}\phi(t,y(t)) = 0 \tag{15}$$

Remark 1.4.1. Actually, 'we can solve' doesn't mean that we can get explict solution. It just becoma a equation without differential term. But with the aid of computer, we can solve its numer with any desired accuracy.

Consider:

$$\frac{\mathrm{d}}{\mathrm{d}t}\phi(t,y) = \frac{\partial\phi}{\partial t} + \frac{\partial\phi}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t} \tag{16}$$

One natural question is that for all $M(t,y), N(t,y)$, does there exist ϕ s.t.

$$M = \frac{\partial\phi}{\partial t} \quad N = \frac{\partial\phi}{\partial y} \tag{17}$$

The answer is **NO**.

Theorem 1.4.2. *Let $M(t,y)$ and $N(t,y)$ be continuous and have continuous partial derivatives with respect to t and y in the rectangle R consisting of those points (t,y) with $a < t < b$ and $c < y < d$. There exists a function $\phi(t,y)$ such that*

$$M(t,y) = \frac{\partial\phi}{\partial t} \quad \text{and} \quad N(t,y) = \frac{\partial\phi}{\partial y}$$

if, and only if,

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}$$

in R.

Proof: 4.3

Definition 1.4.1 (Exact equations)**.**
The differential equation

$$M(t,y) + N(t,y)\frac{dy}{dt} = 0$$

is said to be exact *if*

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}.$$

For exact equations, we can write:

$$\phi(t,y) = \int M(t,y) + \int \Big(N(t,y) - \int \frac{\partial M(t,y)}{\partial y} dt\Big) dy \tag{18}$$

Remark 1.4.3. It's not necessary that $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}$ in a rectangle R as theorem 1.4.2 states. It can be any region without 'hole' inside.

Remark 1.4.4. For differential equation $dy/dt - f(t,y)$, it can be written in the form $M(t,y) + N(t,y)(dy/dt) = 0$ by setting $M(t,y) = -f(t,y)$ and $N(t,y) = 1$.

Now we've solved Exact equations. Suppose

$$M(t,y) + N(t,y)\frac{\mathrm{d}y}{\mathrm{d}t} = 0 \tag{19}$$

is not exact. Could we solve it?
A natural idea is to multiply both terms with a function $\mu(t,y)$:

$$M(t,y)\mu(t,y) + \mu(t,y)N(t,y)\frac{\mathrm{d}y}{\mathrm{d}t} = 0 \tag{20}$$

And if μ is well-chosen, it could become an exact equation, and we can have

$$\frac{\partial}{\partial y}\Big(M(t,y)\mu(t,y)\Big) = \frac{\partial}{\partial t}\Big(N(t,y)\mu(t,y)\Big) \tag{21}$$

Definition 1.4.2 (Integrating Factor)**.**
Any function $\mu(t,y)$ satisfying (21) is called integrating factor for the equation (19)

Then we try to find μ.

$$\frac{\partial}{\partial y}\Big(M(t,y)\mu(t,y)\Big) = \frac{\partial}{\partial t}\Big(N(t,y)\mu(t,y)\Big)$$

$$\Downarrow$$

$$M\frac{\mathrm{d}}{\mathrm{d}y}\mu + \mu\frac{\mathrm{d}}{\mathrm{d}y}M = N\frac{\mathrm{d}\mu}{\mathrm{d}t} + \mu\frac{\mathrm{d}}{\mathrm{d}t}N$$

$$\Downarrow$$

$$-N\frac{\mathrm{d}\mu}{\mathrm{d}t} + M\frac{\mathrm{d}\mu}{\mathrm{d}y} = (\frac{\mathrm{d}N}{\mathrm{d}t} - \frac{\mathrm{d}M}{\mathrm{d}y})\mu$$

Solving a general $\mu(t,y)$ is almost as difficult as solving the ODE equation.

So if μ could only depend on t or y, we could be able to solve it. **It's just an assmuption. We cannot say that there must exists $\mu = \mu(t)$ or $\mu = \mu(y)$. Just for some special cases, if we calculate something and then we find that we are lucky enough.**

4.1 $\mu = \mu(t)$

If μ depends only on t, then $\partial\mu/\partial y = 0$. Therefore,

$$-N\frac{d\mu}{dt} = \left(\frac{\partial N}{\partial t} - \frac{\partial M}{\partial y}\right)\mu,$$

and hence

$$\frac{1}{\mu}\frac{d\mu}{dt} = \frac{\dfrac{\partial M}{\partial y} - \dfrac{\partial N}{\partial t}}{N}. \tag{22}$$

Since the left-hand side depends only on t, the expression on the right-hand side must also be a function of t alone. If

$$\frac{\dfrac{\partial M}{\partial y} - \dfrac{\partial N}{\partial t}}{N} = f(t),$$

then integrating (22) gives

$$\boxed{\mu(t) = C\exp\left(\int f(t)\,dt\right)}, \qquad C \neq 0. \tag{23}$$

4.2 $\mu = \mu(y)$

If μ depends only on y, then $\partial\mu/\partial t = 0$. Therefore,

$$M\frac{d\mu}{dy} = \left(\frac{\partial N}{\partial t} - \frac{\partial M}{\partial y}\right)\mu,$$

and hence

$$\frac{1}{\mu}\frac{d\mu}{dy} = \frac{\dfrac{\partial N}{\partial t} - \dfrac{\partial M}{\partial y}}{M}. \tag{24}$$

Since the left-hand side depends only on y, the expression on the right-hand side must also be a function of y alone. If

$$\frac{\dfrac{\partial N}{\partial t} - \dfrac{\partial M}{\partial y}}{M} = g(y),$$

then integrating (24) gives

$$\boxed{\mu(y) = C\exp\left(\int g(y)\,dy\right)}, \qquad C \neq 0. \tag{25}$$

4.3 Proof of theorem1.4.2

Proof. Fix $t_0 \in (a, b)$. First, suppose that ϕ satisfies

$$\frac{\partial \phi}{\partial t} = M(t, y).$$

Integrating with respect to t, while regarding y as a constant, gives

$$\phi(t, y) = \int_{t_0}^{t} M(s, y)\, ds + h(y), \tag{26}$$

where $h(y)$ is an arbitrary function depending only on y.
Differentiating (26) with respect to y, we obtain

$$\frac{\partial \phi}{\partial y} = \int_{t_0}^{t} \frac{\partial M}{\partial y}(s, y)\, ds + \frac{dh}{dy}(y).$$

We also require $\partial\phi/\partial y = N(t, y)$. Therefore,

$$\frac{dh}{dy}(y) = N(t, y) - \int_{t_0}^{t} \frac{\partial M}{\partial y}(s, y)\, ds. \tag{27}$$

The left-hand side of (27) depends only on y, so the right-hand side must also be independent of t. Differentiating the right-hand side with respect to t, we find

$$\frac{\partial}{\partial t}\left(N(t, y) - \int_{t_0}^{t} \frac{\partial M}{\partial y}(s, y)\, ds\right) = \frac{\partial N}{\partial t}(t, y) - \frac{\partial M}{\partial y}(t, y) = 0$$

Thus, if such a function ϕ exists, then necessarily

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}.$$

Conversely, if this equality holds throughout R, then the right-hand side of (27) has zero derivative with respect to t, and hence depends only on y. We can therefore integrate it with respect to y to obtain a function $h(y)$. Substituting this h into (26) gives a function satisfying

$$\frac{\partial \phi}{\partial t} = M(t, y) \quad \text{and} \quad \frac{\partial \phi}{\partial y} = N(t, y).$$

Consequently, such a function ϕ exists if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}.$$

□

5 The existence-uniqueness theorem: Picard iteration

Many first order differential equations (actually, most) we cannot get the explict solution. However, can we prove that there exists a solution? If so, could we prove that it's the only sulution, or it's the only solution under some conditions?

Consider

$$\frac{\mathrm{d}y}{\mathrm{d}t} = f(t, y), \quad y(t_0) = y_0 \tag{28}$$

So

$$y(t) = y_0 + \int_{t_0}^{t} f(s, y(s))ds \quad \text{as } y(t) \text{ is continuous and } y(t_0) = y_0 \tag{29}$$

Define an operator[2]

$$\mathcal{L}[y](t) = y_0 + \int_{t_0}^{t} f(s, y(s))ds$$

$$\mathcal{L} : C([0, T]) \to C([0, T])$$

For equation

$$y(t) = \mathcal{L}[y](t)$$

we call it *fixed point* of $\mathcal{L}$.
So the problem: Finding the solution of ODE $\to$ Finding the fixed point of $\mathcal{L}$. We'll use a sequence of $\{y_n\}$ to approximate.

In the following subsubsections, we will learn how to construct a sequence of $\{y_n\}$ in 5.1. We'll talk about whether the sequence will converge in 5.2 (But we don't care where it converges). We'll prove that the convergence result is the solution of (29) in 5.3. Last, we'll prove the uniqueness of the solution in 5.4

5.1 Construction of Picard iterates

To solve

$$y(t) = y_0 + \int_{t_0}^{t} f(s, y(s))ds \quad \text{as } y(t) \text{ is continuous and } y(t_0) = y_0 \tag{30}$$

we create a sequance of $\{y(t)\}$ to approximate the answer. We start with the constant function

$$y_0(t) = y_0.$$

Substituting this initial guess into the right-hand side of the integral equation gives the first Picard iterate:

$$y_1(t) = \mathcal{L}[y_0](t) = y_0 + \int_{t_0}^{t} f(s, y_0(s))\, ds = y_0 + \int_{t_0}^{t} f(s, y_0)\, ds.$$

[2]Actually I still don't get fully understand why we do so complex here. It may have some connection with the space? Anyway, see it, accept it, and see what's going on later.

If $y_1(t) = y_0(t)$, then y_0 is already a fixed point of $\mathcal{L}$, so it is a solution of the initial value problem. If not, we substitute y_1 into $\mathcal{L}$ again and obtain

$$y_2(t) = \mathcal{L}[y_1](t) = y_0 + \int_{t_0}^{t} f(s, y_1(s))\, ds.$$

We continue this process. At each step, if

$$y_{n+1}(t) = y_n(t),$$

then y_n is a fixed point of $\mathcal{L}$ and hence a solution. Otherwise, we use y_{n+1} to construct the next iterate. In general, the Picard iterates are defined by

$$\boxed{y_0(t) = y_0, \qquad y_{n+1}(t) = y_0 + \int_{t_0}^{t} f(s, y_n(s))\, ds, \quad n \geq 0.} \tag{31}$$

Notice that every iterate satisfies $y_n(t_0) = y_0$.
Then we'll show the convergence of Picard iterates.

5.2 Convergence of Picard iterates

Lemma 1.5.1. *Choose any two positive numbers a and b, and let R be the rectangle:*

$$t_0 \leq t \leq t_0 + a, \quad |y - y_0| \leq b.$$

Compute

$$M = \max_{(t,y)\in R} |f(t, y)|, \quad \textit{and set } \alpha = \min\left(a, \frac{b}{M}\right).$$

Then,

$$|y_n(t) - y_0| \leq M(t - t_0)$$

for $t_0 \leq t \leq t_0 + \alpha$.

Proof. For $n = 0$,

$$|y_0(t) - y_0| = 0 \leq M(t - t_0).$$

Suppose the result is true for $n = j$. Since $t - t_0 \leq \alpha$,

$$|y_j(t) - y_0| \leq M(t - t_0) \leq M\alpha \leq b.$$

Thus, $(t, y_j(t)) \in R$, so $|f(t, y_j(t))| \leq M$. Therefore,

$$\begin{aligned} |y_{j+1}(t) - y_0| &= \left| \int_{t_0}^{t} f(s, y_j(s))\, ds \right| \\ &\leq \int_{t_0}^{t} |f(s, y_j(s))|\, ds \\ &\leq M(t - t_0). \end{aligned}$$

Hence the result follows for every $n \geq 0$ by induction. $\square$

We now show the convergence of $\{y_n\}$. First, we write y_n in this form:

$$y_n(t) = y_0(t) + [y_1(t) - y_0(t)] + \cdots + [y_n(t) - y_{n-1}(t)]$$

So the sequence $y_n(t)$ converges iff the infinite series converges:

$$[y_1(t) - y_0(t)] + [y_2(t) - y_1(t)] + \cdots + [y_n(t) - y_{n-1}(t)] + \cdots$$

Then, we try to prove a stronger result:

$$\sum_{n=1}^{\infty} |y_n(t) - y_{n-1}(t)| < \infty \Rightarrow \sum_{n=1}^{\infty} \big[y_n(t) - y_{n-1}(t)\big] \text{ converges.} \tag{32}$$

This is not obvious (at least not obvious for me). The result used in (32) will be proved in 5.5.

$$L = \max_{(t,y)\in R} \left| \frac{\partial f}{\partial y}(t,y) \right|.$$

By the mean value theorem, for some $\xi_n(s)$ between $y_n(s)$ and $y_{n-1}(s)$,

$$\begin{aligned} |f(s, y_n(s)) - f(s, y_{n-1}(s))| &= \left| \frac{\partial f}{\partial y}(s, \xi_n(s)) \right| |y_n(s) - y_{n-1}(s)| \\ &\le L |y_n(s) - y_{n-1}(s)|. \end{aligned}$$

where

$$L = \max_{(t,y)\in R} \left| \frac{\partial f}{\partial y}(t,y) \right|.$$

First,

$$\begin{aligned} |y_1(t) - y_0(t)| &= \left| \int_{t_0}^{t} f(s, y_0)\, ds \right| \\ &\le M(t - t_0). \end{aligned}$$

Then

$$\begin{aligned} |y_2(t) - y_1(t)| &= \left| \int_{t_0}^{t} \big[f(s, y_1(s)) - f(s, y_0(s))\big]\, ds \right| \\ &\le L \int_{t_0}^{t} |y_1(s) - y_0(s)|\, ds \\ &\le ML \int_{t_0}^{t} (s - t_0)\, ds \\ &= \frac{ML(t - t_0)^2}{2!}, \end{aligned}$$

and

$$\begin{aligned}|y_3(t) - y_2(t)| &\leq L \int_{t_0}^{t} |y_2(s) - y_1(s)|\, ds \\ &\leq \frac{ML^2(t-t_0)^3}{3!}.\end{aligned}$$

Continuing in the same way,

$$\boxed{|y_n(t) - y_{n-1}(t)| \leq \frac{ML^{n-1}(t-t_0)^n}{n!}, \qquad n \geq 1.} \tag{33}$$

If $L = 0$, then $y_2 = y_1$. For $L > 0$ and $t_0 \leq t \leq t_0 + \alpha$,

$$\begin{aligned}\sum_{n=1}^{\infty} |y_n(t) - y_{n-1}(t)| &\leq \sum_{n=1}^{\infty} \frac{ML^{n-1}(t-t_0)^n}{n!} \\ &= \frac{M}{L} \sum_{n=1}^{\infty} \frac{[L(t-t_0)]^n}{n!} \\ &= \frac{M}{L}\left(e^{L(t-t_0)} - 1\right) \\ &\leq \boxed{\frac{M}{L}\left(e^{\alpha L} - 1\right)} < \infty.\end{aligned}$$

5.3 Is Picard iterates right

Till now, we've proved that $\{y_n\}$ converges. However, we still don't know whether $\lim_{n\to\infty} y_n(t)$ is the answer of (29).
Consider $\lim_{n\to\infty} y_n(t) = y$, and we take limit of both sides of:

$$y_{n+1}(t) = y_0 + \int_{t_0}^{t} f(s, y_n(s))\, ds$$

Then we get

$$y = y_0 + \lim_{n\to\infty} \int_{t_0}^{t} f(s, y_n(s))\, ds \tag{34}$$

Then what we need to do is to justify the right hand side equals to $\int_{t_0}^{t} f(s, y(s))$.
One could also understand that the limit could *pass* the integral, as:

$$\lim_{n\to\infty} \int_{t_0}^{t} f(s, y_n(s))\, ds = \int_{t_0}^{t} \left(\lim_{n\to\infty} f(s, y_n(s))\right) ds = \int_{t_0}^{t} f(s, y(s))\, ds$$

We still need to justify this step. For $m > n$, by (33),

$$\begin{aligned}
|y_m(t) - y_n(t)| &\le \sum_{k=n+1}^{m} |y_k(t) - y_{k-1}(t)| \\
&\le \sum_{k=n+1}^{m} \frac{ML^{k-1}(t-t_0)^k}{k!} \\
&\le \sum_{k=n+1}^{m} \frac{ML^{k-1}\alpha^k}{k!}.
\end{aligned}$$

Let $m \to \infty$. Then

$$\sup_{t_0 \le t \le t_0+\alpha} |y(t) - y_n(t)| \le \sum_{k=n+1}^{\infty} \frac{ML^{k-1}\alpha^k}{k!} \longrightarrow 0. \tag{35}$$

Thus,

$$y_n \longrightarrow y \quad \text{uniformly on } [t_0, t_0+\alpha].$$

By the mean value theorem,

$$\begin{aligned}
|f(s, y_n(s)) - f(s, y(s))| &\le L|y_n(s) - y(s)| \\
&\le L \sup_{t_0 \le r \le t_0+\alpha} |y_n(r) - y(r)|.
\end{aligned}$$

Therefore,

$$\begin{aligned}
&\left| \int_{t_0}^{t} f(s, y_n(s))\, ds - \int_{t_0}^{t} f(s, y(s))\, ds \right| \\
&\le \int_{t_0}^{t} |f(s, y_n(s)) - f(s, y(s))|\, ds \\
&\le \alpha L \sup_{t_0 \le r \le t_0+\alpha} |y_n(r) - y(r)| \\
&\longrightarrow 0.
\end{aligned}$$

So the limit can pass through the integral:

$$\lim_{n\to\infty} \int_{t_0}^{t} f(s, y_n(s))\, ds = \int_{t_0}^{t} f(s, y(s))\, ds.$$

Taking the limit in the Picard iteration gives

$$\boxed{y(t) = y_0 + \int_{t_0}^{t} f(s, y(s))\, ds.} \tag{36}$$

We also need to show that $y(t)$ is continuous. First,

$$y_0(t) = y_0 \quad \Longrightarrow \quad y_0 \text{ is continuous.}$$

If y_n is continuous, then

$$s \longmapsto f(s, y_n(s)) \text{ is continuous}$$

and

$$y_{n+1}(t) = y_0 + \int_{t_0}^{t} f(s, y_n(s))\, ds \quad \Longrightarrow \quad y_{n+1} \text{ is continuous.}$$

Thus, every y_n is continuous. Fix $t^* \in [t_0, t_0 + \alpha]$. For every $\varepsilon > 0$, the uniform convergence in (35) gives an N such that

$$\sup_{t_0 \le t \le t_0+\alpha} |y(t) - y_N(t)| < \frac{\varepsilon}{3}.$$

Since y_N is continuous, there exists $\delta > 0$ such that

$$|t - t^*| < \delta \quad \Longrightarrow \quad |y_N(t) - y_N(t^*)| < \frac{\varepsilon}{3}.$$

Therefore,

$$\begin{aligned} |y(t) - y(t^*)| &\le |y(t) - y_N(t)| + |y_N(t) - y_N(t^*)| + |y_N(t^*) - y(t^*)| \\ &< \varepsilon. \end{aligned}$$

Hence,

$$\boxed{y \in C([t_0, t_0 + \alpha]).} \tag{37}$$

Finally,

$$y(t_0) = y_0, \qquad \frac{dy}{dt} = f(t, y(t)).$$

Thus, $y(t)$ is a solution of (29).

5.4 Uniqueness

Theorem 1.5.2. *Let f and $\partial f/\partial y$ be continuous in the rectangle*

$$R: \quad t_0 \le t \le t_0 + a, \quad |y - y_0| \le b.$$

Compute

$$M = \max_{(t,y)\in R} |f(t, y)|, \quad \text{and set } \alpha = \min\left(a, \frac{b}{M}\right).$$

Then, the initial-value problem

$$y' = f(t, y), \quad y(t_0) = y_0 \tag{38}$$

has a unique solution $y(t)$ on the interval $t_0 \le t \le t_0 + \alpha$. In other words, if $y(t)$ and $z(t)$ are two solutions of (38), then $y(t)$ must equal $z(t)$ for $t_0 \le t \le t_0 + \alpha$.

Proof. Theorem 2 guarantees the existence of at least one solution $y(t)$ of (38). Suppose that $z(t)$ is a second solution of (38). Then,

$$y(t) = y_0 + \int_{t_0}^{t} f(s, y(s))\, ds \quad \text{and} \quad z(t) = y_0 + \int_{t_0}^{t} f(s, z(s))\, ds.$$

Subtracting these two equations gives

$$|y(t) - z(t)| = \left| \int_{t_0}^{t} [f(s, y(s)) - f(s, z(s))] \, ds \right|$$

$$\leq \int_{t_0}^{t} |f(s, y(s)) - f(s, z(s))| \, ds$$

$$\leq L \int_{t_0}^{t} |y(s) - z(s)| \, ds,$$

where L is the maximum value of $|\partial f / \partial y|$ for (t, y) in R. As Lemma 1.5.3 below shows, this inequality implies that $y(t) = z(t)$. Hence, the initial-value problem (38) has a unique solution $y(t)$. □

Lemma 1.5.3. *Let $w(t)$ be a nonnegative function, with*

$$w(t) \leq L \int_{t_0}^{t} w(s) \, ds.$$

Then, $w(t)$ is identically zero.

Proof. Set

$$U(t) = \int_{t_0}^{t} w(s) \, ds.$$

Then

$$U'(t) = w(t) \leq L \int_{t_0}^{t} w(s) \, ds = LU(t).$$

Therefore,

$$\frac{d}{dt} \left(e^{-L(t-t_0)} U(t) \right) = e^{-L(t-t_0)} \big(U'(t) - LU(t) \big)$$
$$\leq 0.$$

Thus,

$$0 \leq e^{-L(t-t_0)} U(t) \leq U(t_0) = 0,$$

so $U(t) = 0$. Finally,

$$0 \leq w(t) \leq L \int_{t_0}^{t} w(s) \, ds = LU(t) = 0.$$

Hence, $\boxed{w(t) = 0}$ for $t_0 \leq t \leq t_0 + \alpha$. □

5.5 Proof of (32)

First, consider a sequence $\{a_n\}_{n \geq 1} \subset \mathbb{R}$ such that

$$\sum_{n=1}^{\infty} |a_n| < \infty.$$

Define

$$S_n = \sum_{k=1}^{n} a_k, \qquad A_n = \sum_{k=1}^{n} |a_k|.$$

Since $\{A_n\}$ converges, it is a Cauchy sequence. Thus,

$$\forall \varepsilon > 0, \qquad \exists N \in \mathbb{N}, \qquad m > n \geq N$$

$$\Downarrow$$

$$|A_m - A_n| = \sum_{k=n+1}^{m} |a_k| < \varepsilon.$$

Then

$$\begin{aligned} |S_m - S_n| &= \left| \sum_{k=n+1}^{m} a_k \right| \\ &\leq \sum_{k=n+1}^{m} |a_k| \\ &< \varepsilon. \end{aligned}$$

Therefore,

$$\{S_n\} \text{ is Cauchy} \overset{\mathbb{R} \text{ is complete}}{\Longrightarrow} S_n \longrightarrow S \in \mathbb{R},$$

and hence

$$\boxed{\sum_{n=1}^{\infty} |a_n| < \infty \quad \Longrightarrow \quad \sum_{n=1}^{\infty} a_n \text{ converges.}} \tag{39}$$

Chapter 2

Second Order Linear Differential Equations

1 Introduction

We've considered first order differential equations. Now we try to understand second order.
Consider

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = f(t, y, \frac{\mathrm{d}y}{\mathrm{d}t}), \quad y(t_0) = y_0, y'(t_0) = y'_0 \tag{1}$$

Think about how we define linear first order equation before. Now we define second order linear differential equation.

Definition 2.1.1 (Second order linear differential equation)**.**
The equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + p(t)\frac{\mathrm{d}y}{\mathrm{d}t} + q(t)y = g(t) \tag{2}$$

is called second order linear differential equation.

Sadly, almost all second order differential equations are extremely difficult to solve. Till now we can only solve (2)

Consider an easier case of (2) when $g(t) = 0$. Like what we did in forst order, we can also define second order linear homogeneous equation.

Definition 2.1.2 (Second order linear homogeneous equation)**.**
The equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + p(t)\frac{\mathrm{d}y}{\mathrm{d}t} + q(t)y = 0 \tag{3}$$

is called second order linear homogeneous equation.

However, it could be very difficult even in solving second order linear homogeneous equation. We'll first consider whether it actually has a solution (or solutions) and its properties.

2 Algebraic properties of solutions

Consider

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + p(t)\frac{\mathrm{d}y}{\mathrm{d}t} + q(t)y = 0 \tag{4}$$

Then we consider an operator $L[y](t)$:

$$L[y](t) = \frac{\mathrm{d}^2y}{\mathrm{d}t^2} + p(t)\frac{\mathrm{d}y}{\mathrm{d}t} + q(t)y \tag{5}$$

One can understand $L[y](t)$ as 'function on functions'. We input a function $y(t)$, then output a function $L[y](t)$. With the help of operator, we could analysis the solutions of (4) easier.

Property 1 $L[cy] = cL[y]$
Property 2 $L[y_1 + y_2] = L[y_1] + L[y_2]$

Definition 2.2.1 (Linear operator)**.**
An operator L which satisfies properties 1 and 2 is called a linear operator. All other operators are non-linear.

Thus the solution of (4) could be written in the form of (5) as

$$L[y](t) = 0 \tag{6}$$

Note the right hand side is also a function instead a scalar.

For any solution $y(t)$ of (6), $cy(t)$ is also the solution. For any solutions $y_1(t)$, $y_2(t)$ of (6), $c_1y_1(t) + c_2y_2(t)$ is also the solution.

Now we give a theorem which will be proved very later:

Theorem 2.2.1 (Existence–uniqueness Theorem)**.** *Let the functions $p(t)$ and $q(t)$ be continuous in the open interval $\alpha < t < \beta$. Then, there exists one, and only one function $y(t)$ satisfying the differential equation (3) on the entire interval $\alpha < t < \beta$, and the prescribed initial conditions*

$$y(t_0) = y_0, \qquad y'(t_0) = y'_0.$$

In particular, any solution $y = y(t)$ of (3) which satisfies $y(t_0) = 0$ and $y'(t_0) = 0$ at some time $t = t_0$ must be identically zero.

Consider we now obtain y_1 and y_2 as the solutions of the ODE. Could y_1, y_2 span the space of the solutions? Or y_1 and y_2 are the $'same'$? We could use Theorem2.2.2 to solve the problem. Let $y(t)$ be any solution of (4). We must find constants c_1 and c_2 such that $y(t) = c_1y_1(t) + c_2y_2(t)$. To this end, pick a time t_0 in the interval (α, β) and let y_0 and y'_0 denote the values of y and y' at $t = t_0$. The constants c_1 and c_2, if they exist, must satisfy the two equations

$$c_1y_1(t_0) + c_2y_2(t_0) = y_0,$$

$$c_1y'_1(t_0) + c_2y'_2(t_0) = y'_0.$$

Multiplying the first equation by $y_2'(t_0)$, the second equation by $y_2(t_0)$, and subtracting gives

$$c_1\big[y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)\big] = y_0y_2'(t_0) - y_0'y_2(t_0).$$

Similarly, multiplying the first equation by $y_1'(t_0)$, the second equation by $y_1(t_0)$, and subtracting gives

$$c_2\big[y_1'(t_0)y_2(t_0) - y_1(t_0)y_2'(t_0)\big] = y_0y_1'(t_0) - y_0'y_1(t_0).$$

Hence,

$$c_1 = \frac{y_0y_2'(t_0) - y_0'y_2(t_0)}{y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)}$$

and

$$c_2 = \frac{y_0'y_1(t_0) - y_0y_1'(t_0)}{y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)}$$

if $y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0) \neq 0$.

Theorem 2.2.2. *Let $y_1(t)$ and $y_2(t)$ be two solutions of (4) on the interval $\alpha < t < \beta$, with*

$$y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

unequal to zero in this interval. Then,

$$y(t) = c_1y_1(t) + c_2y_2(t)$$

is the general solution of (4).

Definition 2.2.2 (Wronskian)**.**
The quantity

$$y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

is called the Wronskian *of y_1 and y_2, and is denoted by*

$$W(t) = W[y_1, y_2](t).$$

Theorem 2.2.3. *Let $p(t)$ and $q(t)$ be continuous in the interval $\alpha < t < \beta$, and let $y_1(t)$ and $y_2(t)$ be two solutions of (3). Then, $W[y_1, y_2](t)$ is either identically zero, or is never zero, on the interval $\alpha < t < \beta$.*

Lemma 2.2.4. *Let $y_1(t)$ and $y_2(t)$ be two solutions of the linear differential equation*

$$y'' + p(t)y' + q(t)y = 0.$$

Then, their Wronskian

$$W(t) = W[y_1, y_2](t) = y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

satisfies the first-order differential equation

$$W' + p(t)W = 0.$$

2.1 Proof of Theorem 2.2.2

Theorem (2.2.2) Let $y_1(t)$ and $y_2(t)$ be two solutions of (3) on the interval $\alpha < t < \beta$, with

$$y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

unequal to zero in this interval. Then,

$$y(t) = c_1y_1(t) + c_2y_2(t)$$

is the general solution of (4).

Proof. Let $y(t)$ be any solution of (4). We must find constants c_1 and c_2 such that $y(t) = c_1y_1(t) + c_2y_2(t)$. To this end, pick a time t_0 in the interval (α, β) and let y_0 and y_0' denote the values of y and y' at $t = t_0$. The constants c_1 and c_2, if they exist, must satisfy the two equations

$$c_1y_1(t_0) + c_2y_2(t_0) = y_0,$$

$$c_1y_1'(t_0) + c_2y_2'(t_0) = y_0'.$$

Multiplying the first equation by $y_2'(t_0)$, the second equation by $y_2(t_0)$, and subtracting gives

$$c_1\left[y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)\right] = y_0y_2'(t_0) - y_0'y_2(t_0).$$

Similarly, multiplying the first equation by $y_1'(t_0)$, the second equation by $y_1(t_0)$, and subtracting gives

$$c_2\left[y_1'(t_0)y_2(t_0) - y_1(t_0)y_2'(t_0)\right] = y_0y_1'(t_0) - y_0'y_1(t_0).$$

Hence,

$$c_1 = \frac{y_0y_2'(t_0) - y_0'y_2(t_0)}{y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)}$$

and

$$c_2 = \frac{y_0'y_1(t_0) - y_0y_1'(t_0)}{y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)}$$

if $y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0) \neq 0$.
Now, let

$$\phi(t) = c_1y_1(t) + c_2y_2(t)$$

for this choice of constants c_1, c_2. We know that $\phi(t)$ satisfies (4), since it is a linear combination of solutions of (4). Moreover, by construction, $\phi(t_0) = y_0$ and $\phi'(t_0) = y_0'$. Thus, $y(t)$ and $\phi(t)$ satisfy the same second-order linear homogeneous equation and the same initial conditions. Therefore, by the uniqueness part of Theorem 2.2.1, $y(t)$ must be identically equal to $\phi(t)$; that is,

$$y(t) = c_1y_1(t) + c_2y_2(t), \quad \alpha < t < \beta.$$

Theorem 2 is an extremely useful theorem since it reduces the problem of finding all solutions of (4), of which there are infinitely many, to the much simpler problem of finding just two solutions $y_1(t), y_2(t)$. The only condition imposed on the solutions

$y_1(t)$ and $y_2(t)$ is that the quantity

$$y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

be unequal to zero for $\alpha < t < \beta$. When this is the case, we say that $y_1(t)$ and $y_2(t)$ are a *fundamental set of solutions* of (4), since all other solutions of (4) can be obtained by taking linear combinations of $y_1(t)$ and $y_2(t)$. $\square$